CONFIDENTIAL · TECHNICAL ASSESSMENT

Path+ Infrastructure Capacity & Scalability Assessment

Evidence-based capacity plan for supporting 100,000 registered users with normal daily social networking activity.

PREPARED FOR
Path+ Product & Engineering Leadership

ASSESSMENT DATE
2026-11-01

SCOPE
Backend, database, storage, VPS, CDN, cost, security

CLASSIFICATION
Analysis only — no application code was modified

Authoritative backend inspected: C:\Users\admin\Downloads\new\path-plus-main\path-plus-main\backend
Mobile workspace inspected: C:\Users\admin\Downloads\path-plus-main\path-plus-main\mobile
Legend used throughout: Fact from source · Assumption modeled · Unknown needs measurement

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> **Important:** No production usage telemetry was available during this assessment. Workload figures are *modeled* from published social-app heuristics and Path+ feature surface area. They are not measured production statistics.

1 Executive Summary

Path+ can reach 100,000 registered users, but **not on the current Free-tier Supabase + single 512 MB Bun process configuration** without material reliability and performance risk. The limiting factors are database/API fan-out patterns and media egress—not raw VPS CPU alone.

MODELED DAU (NORMAL)	PEAK API RPS (NORMAL)	RECOMMENDED MONTHLY OPEX
40,000	**666.7**	**~$650–1,400**
40% of 100k registered · assumption	≈5× average · assumption	After media optimization; egress is the swing factor

Verdict

- Fact Backend is a single Bun/Hono process (PM2 fork, 512 MB restart limit) behind nginx on a VPS; data plane is Supabase Postgres + Storage + Realtime + Auth.
- Fact Mobile client disables Supabase image transforms (`IMAGE_TRANSFORMS_ENABLED = false`) because the current tenant returns FeatureNotEnabled—full-size media egress today.
- Assumption At Normal usage, ~11,520,000 API requests/day and ~23,065 GB/month combined transfer.
- **Supabase Free Plan is not suitable** for this target (500 MB DB, 1 GB storage, 200 realtime connections, pausing risk, no PITR).
- Stay on Supabase (upgrade compute + Pro/Team). Add Redis, CDN/object-storage egress control, and fix P0 code bottlenecks before or while scaling users.

Most critical scalability risks discovered

1. **P0 — Shared rate-limit key** (`src/lib/rate-limit.ts`): Authorization header sliced to 48 chars likely collapses authenticated users into one global 400 req/min bucket.
2. **P0 — Feed / inbox query fan-out**: hot paths issue ~10–25 Supabase HTTP round-trips per request; inbox unread COUNTs scale with conversations.
3. **P0/P1 — Unfiltered global messages realtime subscription** on mobile amplifies Realtime fan-out.
4. **P1 — Single process + in-memory caches + base64 uploads**: cannot horizontally scale safely; 512 MB restart under concurrent uploads.
5. **P1 — Full-size media without transforms/CDN strategy**: egress and latency dominate cost before API CPU does.

2 Existing Architecture Assessment

Path+ is a production iOS social network (Saudi Arabia focus, worldwide availability) with registration, profiles, moments (photo/video/text), chronological feed, friendships, DMs, reactions/comments, push, reporting/blocking, and an admin dashboard.

As-deployed topology (from repository evidence)

Layer	Current implementation	Evidence
Client	Expo / React Native iOS app → HTTPS API + direct Supabase Auth/Storage/Realtime	mobile `src/lib/api.ts`, `supabase.ts`
Edge	nginx TLS termination; hosts api / admin / site.pathplus.store	`deploy/nginx.pathplus.conf`
API	Bun + Hono on port 3000; PM2 app `pathplus-api`; 1 fork instance; max_memory_restart 512M	`ecosystem.config.cjs`
Admin	Next.js standalone on port 3001; PM2 `pathplus-admin`	same
Database	Supabase Postgres via `@supabase/supabase-js` (PostgREST). No app-level pool.	`package.json`, routes
Auth	Supabase Auth (Bearer JWT). Custom OTP signup/reset via Resend. Separate admin JWT (jose).	`routes/auth.ts`, admin auth
Realtime	No WebSocket server on VPS. Mobile uses Supabase Realtime `postgres_changes`.	mobile hooks
Media	Supabase Storage buckets `Avatars`, `Covers`, `Posts` (public). Limits 15/50/25 MB.	`routes/upload.ts`
Push	Expo Push API → APNs/FCM; tokens in `user_devices`	`lib/push.ts`
Cache / queue	None (Redis/workers absent). In-process Maps + setInterval crons.	grep / package.json

> **Hostinger:** User context states Hostinger VPS hosting. Repository confirms generic Linux VPS paths (`/root/.bun/bin/bun`, nginx, PM2) but does *not* embed Hostinger plan SKUs. Exact current vCPU/RAM is `Unknown` and should be confirmed on the VPS.

> The authoritative backend path contains nested legacy copies (`backend/backend/`, `backend/Path-Plus/`). Live runtime code is `backend/src`. Treat nested trees as archival risk, not production.

3 Project Technology Stack

Backend (authoritative path)

- Runtime: Bun
- Framework: Hono ^4.12.18
- Validation: Zod 4.1.11
- DB client: @supabase/supabase-js ^2.101.1
- DDL helper: postgres ^3.4.9 (max:1)
- Admin JWT: jose; passwords: bcryptjs
- Email: Resend
- Rate limit: hono-rate-limiter (memory store)
- Admin UI: Next.js 15 standalone
- Prisma/Better Auth: stubbed / unused at runtime

Mobile (inspected workspace)

- Expo ~58 / RN 0.88 (workspace; docs partially stale)
- React Query 5; Zustand; FlashList
- Supabase JS for Auth/Storage/Realtime
- expo-notifications (Expo push tokens)
- Client JPEG resize (~1280px, q≈0.8)
- Video picker H264 960×540, ≤50 MB / 60s
- API base: `EXPO_PUBLIC_BACKEND_URL` → api.pathplus.store

Core API surface (mounted)

/api/auth · /api/posts · /api/friends · /api/notifications · /api/conversations · /api/upload · /api/social · /api/admin · moderation & users routers · health endpoints

Environment variable names (no secret values)

SUPABASE_URL, SUPABASE_ANON_KEY, SUPABASE_SERVICE_ROLE_KEY, DATABASE_URL, PORT, BACKEND_URL, PUBLIC_APP_URL, RESEND_API_KEY, RESEND_FROM_EMAIL, ADMIN_JWT_SECRET, ADMIN_JWT_EXPIRES_IN, ADMIN_CORS_ORIGIN, ADMIN_DEFAULT_PASSWORD, CONFIG_ENCRYPTION_KEY, GOOGLE_PLACES_API_KEY, OPENAI_API_KEY, BETTER_AUTH_SECRET, EXPO_ACCESS_TOKEN; mobile EXPO_PUBLIC_BACKEND_URL, EXPO_PUBLIC_SUPABASE_URL, EXPO_PUBLIC_SUPABASE_ANON_KEY.

4 Capacity Assumptions & Methodology

Inputs from the brief (accepted)

- Registered users: **100,000**
- DAU band: 30%–50% → 30k–50k
- Sessions/active user/day: 4–8
- Session duration: 5–15 minutes
- Geography: Saudi Arabia heavy + international

Additional modeling assumptions (explicit)

Parameter	Conservative	Normal	Heavy	Rationale
DAU %	30%	40%	50%	Brief band
Sessions / DAU	4	6	8	Brief band
Session minutes	6	10	14	Within 5–15
API calls / session	28	48	75	Feed, badges, chat, RT invalidations observed in client
DB RTTs / API call	8	10	14	Current fan-out architecture (feed often higher)
Peak multiplier	4×	5×	6×	Evening KSA peak heuristic
Photo size	~420 KB JPEG after client compress			Client manipulator settings
Video size	~9.5 MB average retained			960×540 export; ≤50 MB cap

Calculation method

Daily API = DAU × sessions × API/session. Avg RPS = daily API / 86,400. Peak RPS = Avg × peak factor. Avg concurrent users ≈ DAU × sessions × session_min / (24×60). Peak concurrent ≈ Avg × peak factor. Realtime channels ≈ peak concurrent × ~4.2–5 channels/user (tabs layout subscriptions + chat).

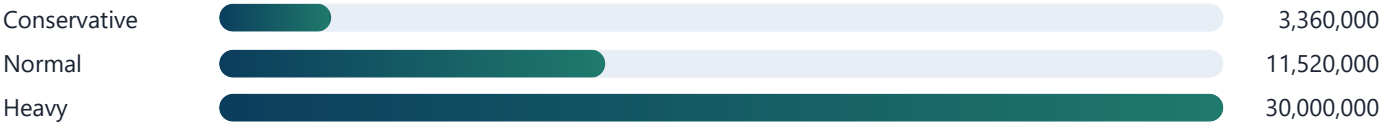
> Where Path+ source shows heavier multipliers (e.g., 15–25 RTTs on feed), Normal/Heavy DB figures intentionally sit above a "clean" 2–3 RTT design to reflect *current* code cost.

5 Three Workload Scenarios

All figures below are Assumption modeled outputs for 100,000 registered users—not measured telemetry.

Metric	Conservative	Normal	Heavy
Daily active users	30,000	40,000	50,000
Daily sessions	4	6	8
Daily API requests	3,360,000	11,520,000	30,000,000
Average API RPS	38.9	133.3	347.2
Peak API RPS	155.6	666.7	2,083.3
Peak concurrent users	2,000	8,333	23,333
Peak concurrent API in-flight	54	233	729
Peak realtime channels	8,400	37,500	116,667
Daily DB round-trips (modeled)	26,880,000	115,200,000	420,000,000
Daily DB reads	22,041,600	94,464,000	344,400,000
Daily DB writes	4,838,400	20,736,000	75,600,000
Daily messages	32,400	134,400	440,000
Daily notifications	120,000	320,000	700,000
Daily photo uploads	1,800	5,120	13,200
Daily video uploads	300	880	2,600
Daily photo downloads/views	540,000	1,120,000	2,250,000
Daily video plays (starts)	45,000	120,000	300,000
Monthly total bandwidth (GB)	7,268	23,065	81,879
Daily media growth (GB)	3.50	10.21	29.41
Daily DB growth (MB)	77.9	244.3	651.6

Daily API volume comparison



Interpretation for Path+

- **Conservative** (~156 peak RPS): achievable on a modest VPS *after* P0 fixes; Free Supabase still inadequate.
- **Normal** (~667 peak RPS): design target for this report. Needs Pro Supabase with Large/XL compute class, Redis, CDN, and query consolidation.

- **Heavy** (~2,083 peak RPS): requires multi-instance API, stronger DB (XL/2XL), realtime connection overages, and media CDN offload.

6 Database Capacity Analysis

Engine & hosting

Fact PostgreSQL on Supabase, accessed exclusively through PostgREST/supabase-js (service role + user JWT clients). Prisma schema is an unused SQLite stub. No application connection pool; concurrency bounded by Supabase PostgREST / Supavisor limits.

Schema coverage (from migrations + boot SQL)

profiles, posts, reactions, comments, friendships, notifications, conversations, conversation_participants, messages, close_friends, user_blocks, saved_posts, hashtags, post_hashtags, mentions, post_shares, post_views, reports, user_devices, account_deletion_requests, pending_registrations, password_reset_otps, admin_*, app_settings/content, external_service_settings. Storage buckets: Avatars, Covers, Posts.

Indexes exist on major foreign keys and time-ordered access paths (posts, messages, notifications, friendships). Likely gaps: pg_trgm for ILIKE search; composite keys matching ORDER BY created_at,id; notification dedupe composites.

Expected records at 100k users (modeled)

Table group	Expected rows	Notes
profiles / auth.users	100,000	1:1 with registered users
posts	1.8M – 4.5M	0.5–1.2 posts/user/month × 12–24 mo accumulation
reactions	12M – 40M	High-volume; unique (post,user,type)
comments	2M – 8M	Social interaction growth
friendships	1.5M – 4M	~15–40 friends average
conversations	400k – 1.2M	1:1 dominant
messages	25M – 120M	Fastest growing chat table
notifications	40M – 150M	Requires retention policy
user_devices	120k – 200k	Multi-device tokens
reports / blocks	50k – 250k	Moderation volume
hashtags / post_hashtags	200k – 2M	Search & discovery
admin_* / settings	< 50k	Low volume

Growth forecast (Normal daily growth model)

Horizon	DB data	Indexes (~35%)	Object storage	Backup store
3 months	33.5 GB	11.7 GB	1.10 TB	73.6 GB
6 months	54.9 GB	19.2 GB	2.02 TB	120.9 GB
12 months	97.9 GB	34.3 GB	3.86 TB	215.3 GB
24 months	183.7 GB	64.3 GB	7.53 TB	404.2 GB

Baseline at “day 0 of 100k steady state” modeled as ~12 GB DB + ~180 GB media — Assumption , replace with Supabase dashboard measurements.

Free plan suitability

> **Supabase Free is not suitable** for 100k users: 500 MB DB, 1 GB file storage, 5 GB egress, 200 peak realtime connections, 50k MAU, project pausing, no automatic backups/PITR. Path+ already uses Auth + Storage + Realtime + heavy PostgREST traffic.

Recommended Postgres sizing (100k, Normal)

Item	Recommendation	Why
Plan	Supabase Pro (or Team if compliance)	Backups, MAU headroom, Smart CDN, no pause
Compute	Large (2 vCPU / 8 GB) → XL (4 / 16 GB) if feed RPCs not added	Fan-out + realtime RLS CPU
Disk	Start 40–80 GB GP, monitor	Data + indexes + bloat
Pooler connections	Keep service-role chattiness down; prefer SQL RPCs	App opens many short HTTP queries today
PITR	Enable 7–14 day PITR	Social data RPO minutes–hours
Retention	Notifications 90 days; message media lifecycle; admin_logs 180 days	Contain fastest tables

Independent Postgres (e.g., managed RDS/Cloud SQL) is optional later; migrating off Supabase is *not* recommended now because Auth, Realtime, and Storage are deeply integrated. Cost delta does not justify rewrite risk at this stage.

7 VPS Server Capacity Analysis

Fact Process model: one Bun API + one Next admin, both 512 MB PM2 limits, fork mode. Marketing HTML also served from the API process. Uploads may buffer base64 in memory. No Redis. Boot runs ~190 exec_sql statements + backfills.

Three infrastructure configurations

Config	Compute	Roles	When
1. Minimum viable production	Hostinger KVM 4 — 4 vCPU / 16 GB / 200 GB NVMe / 16 TB BW Intro ~$12.99/mo · renews ~$28.99/mo (verified Hostinger public pricing, 24-mo term)	API+admin+nginx only. DB/storage remain on Supabase. Raise PM2 memory to 1–1.5 GB. Fix P0 rate-limit & boot jobs.	After code hotfixes; ≤~10–30k users transitioning up
2. Recommended production (100k)	Hostinger KVM 8 — 8 vCPU / 32 GB / 400 GB / 32 TB Intro ~$25.99/mo · renews ~$49.99/mo or 2× KVM 4 (API primary + worker/admin)	2 Bun instances behind nginx upstream once Redis added; separate worker for push/cron; admin on same or second box.	50k–100k with Normal load
3. High availability	2–3 VPS (API) + managed Redis + optional staging VPS; multi-AZ mindset	Active-active API, health-checked LB (Cloudflare/nginx), isolated workers, offsite backups	>100k or stricter uptime

Resource rationale

- **CPU:** Bun is single-threaded per process; PNG avatar generation + bcryptjs + JSON media inflate compete on one core today → need multiple processes, not only bigger RAM.
- **RAM:** 512 MB restart limit is unsafe for 50 MB video base64 paths; budget ≥2 GB RSS headroom per API instance after fixes.
- **Disk:** App+logs only on VPS (50–100 GB fine). Do not store user media on VPS.
- **Bandwidth:** Hostinger 16–32 TB is ample for API JSON; media must not hairpin through VPS.

> 4 vCPU/16 GB is a valid *API host* size only if DB/media/realtime stay off-box and P0/P1 code issues are fixed. It is insufficient as a "everything on one VPS including Postgres/media" design—and Path+ already correctly externalizes DB/media to Supabase.

8 Image & Video Storage Analysis

Question	Finding
Where are photos stored?	Fact Supabase Storage public buckets Avatars, Covers, Posts
Where are videos stored?	Fact Same Posts bucket; client often uploads direct to Supabase
CDN?	Supabase Smart CDN on paid plans; no separate Cloudflare/R2 config in repo. Image transforms disabled in app.
Served from VPS?	Only default generated avatars under `/static/default-avatars`
Compression?	Client-side JPEG resize; no server sharp/ffmpeg
Transcoding / multi-bitrate?	Fact None
Thumbnails?	Not generated server-side

Storage estimate for 100k users (Normal accumulation ~12 months)

Category	Estimate	Basis
Compressed photos	~749 GB / year incremental	5,120 photos/day × 420 KB
Original photos	≈0 (not retained separately)	Client uploads already compressed
Videos (single rendition)	~2,980 GB / year	880/day × 9.5 MB
Transcoded ladder	+0 today; +1.6–2.2× if added	Proposed only
Thumbnails	~49.9 GB / year if generated	28 KB each
Avatars/covers	~15–40 GB cumulative	~180 KB × users × revisions
Temp uploads	<5 GB	Should be ephemeral
Backup/archive of media	+100% cold copy optional	R2 IA / Glacier-class

Using growth table: ~3.86 TB object storage at 12 months and ~7.53 TB at 24 months under Normal model (includes baseline).

Storage platform comparison

Option	Pros	Cons / cost signal
Stay Supabase Storage + enable Image Transforms	Already integrated; Smart CDN; least code change	Egress $0.09/GB uncached / $0.03 cached beyond 250 GB; transforms billed; Free transforms unavailable
Cloudflare R2 + CDN	Egress $0; $0.015/GB-mo storage (verified)	Migration + signed URL upload rewrite; ops ownership
Amazon S3 + CloudFront	Mature ecosystem, lifecycle rules	Egress fees dominate social apps; more complex than R2 for this use case

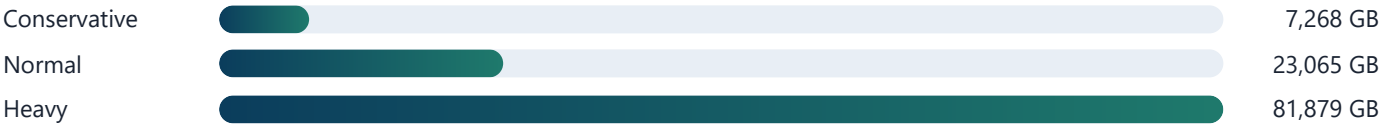
Recommendation: Short term — upgrade Supabase, enable image transformations, turn `IMAGE_TRANSFORMS_ENABLED` on after verifying. Medium term — evaluate R2 for hot media if monthly egress charges exceed ~$150–200.

9 Bandwidth & CDN Analysis

Monthly transfer model (GB). Photo downloads apply a cache-miss factor (device/CDN). Video uses average megabytes transferred per play-start (partial play / `preload`), not full object size every time.

Traffic type	Conservative	Normal	Heavy
API responses	1,730.3 GB	5,932.6 GB	15,449.5 GB
Photo uploads	21.6 GB	61.5 GB	158.6 GB
Video uploads	83.5 GB	244.9 GB	723.6 GB
Photo downloads	2,271.1 GB	5,383.3 GB	14,870.2 GB
Video streaming (partial)	2,900.4 GB	10,546.9 GB	48,339.8 GB
Realtime messaging	1.5 GB	6.2 GB	20.1 GB
Other / overhead	259.6 GB	889.9 GB	2,317.4 GB
TOTAL monthly	**7,268.0 GB**	**23,065.3 GB**	**81,879.4 GB**

Monthly bandwidth comparison



Hostinger VPS bandwidth

KVM 4/8 advertise 16–32 TB/month — sufficient for API JSON and admin traffic. **It is the wrong place to terminate media downloads.** If media were proxied via VPS, Normal/Heavy video+photo traffic would waste CPU/RAM and risk fair-use scrutiny.

Saudi Arabia & international delivery

- Prefer a CDN with Middle East PoPs (Cloudflare or Supabase Smart CDN).
- Keep API region close to DB (Supabase project region). Confirm current region Unknown .
- Adaptive image sizes (avatar 150², feed ~720–1080) cut photo egress dramatically.

Cost reduction levers

1. Enable image transforms / responsive URLs
2. Generate posters/thumbnails; change video `preload` to metadata
3. Optional HLS / dual-resolution later
4. CDN cache immutable object paths
5. Keep timeline cursor pagination (already present on feed)
6. Avoid base64 upload path through API

10 Security, Reliability & Backup

Area	Current state	Target for 100k
DB security	Supabase RLS + service role on server	Audit RLS; least-privilege; rotate service role; restrict exec_sql
API protection	Rate limits present but keying flawed; CORS allows broad localhost/LAN patterns	Per-user limits; tighten CORS; nginx limit_req; WAF
Auth	Supabase JWT; OTP hashed; admin jose JWT + denylist	Confirm admin default password rotated; local JWT verify to cut Auth RTT
DDoS	No Cloudflare/WAF in repo	Cloudflare proxy for api/site
Media uploads	MIME allowlists + size caps; public buckets	Signed upload URLs; private buckets + signed read for DMs if needed
Backups	Depends on Supabase plan — Free has none	Daily + PITR; monthly offsite export
RTO / RPO	Not defined in repo	RTO ≤ 4h API; RPO ≤ 15m DB (PITR)
Monitoring	console + PM2; public health endpoints leak readiness detail	APM, error tracking, uptime checks; auth-protect ready probes

Critical weaknesses observed in code

- Shared authenticated rate-limit bucket (P0)
- `cf-connecting-ip` trusted without Cloudflare → auth limiter bypass
- Unauthenticated `/api/upload/test-storage` bucket listing
- Public `/health?ready=1` exposes configuration flags
- Unlimited CPU path on default avatar generator
- Docs mention default admin credentials historically — verify rotation (Unknown)

11 Performance Bottlenecks & Required Improvements

ID	Issue / location	Impact	Solution	Pri	Type
A	Rate-limit key `authorization.slice(0,48)` — `src/lib/rate-limit.ts`	Global 429s under modest concurrency	Key by verified userId / full token hash	P0	Code
B	Feed fan-out — `routes/posts.ts`, `lib/load-posts.ts`	15–25 Supabase RTTs / feed page	SQL RPC / view; denormalize reaction summaries	P0	Code+DB
C	Inbox unread N+1 — `routes/conversations.ts`	Linear with chats; also on every push badge	last_message + unread counters	P0	Code+DB
D	Global messages realtime — mobile `use-unread-messages.ts`	Realtime fan-out & connection cost	Filter by participant; server badge channel	P0	Code
E	20s friendship poll fallback	Steady background API load	Subscribe-first; backoff poll	P1	Code
F	Boot DDL/backfills — `src/index.ts`	Restart storms hit DB	One-shot migrations; feature flag	P1	Code
G	Push per-token HTTP + receipt polls — `lib/push.ts`	Latency & outbound amplification	Batch 100; queue worker	P1	Code+Infra
H	Single Bun instance / memory caches	No HA; sticky cache	Redis + 2+ instances	P1	Infra+Code
I	Base64 upload buffering — `routes/upload.ts`	OOM / PM2 restart	Signed direct uploads only	P1	Code
J	ILIKE search w/o trigram	Seq scans as tables grow	pg_trgm GIN; use post_hashtags	P2	DB+Code
K	Image transforms off — `lib/images.ts`	Egress & latency	Enable on Pro; or R2+image resizing	P1	Infra+Code
L	Video WebView preload=auto	Bandwidth amplification	metadata preload; virtualize	P2	Code
M	No APM / metrics	Blind scaling	OpenTelemetry + uptime + pg_stat_statements	P2	Infra

12 Monthly Cost Estimation (USD)

Prices marked **Verified** were checked against public provider pages on the assessment date. Usage lines are **Estimates**. Taxes excluded. Hostinger shows intro vs renewal; table uses renewal-oriented planning rates for conservatism on Budget/Recommended where noted.

A. Budget configuration

Service	Plan / size	Monthly	Status
Hostinger VPS	KVM 4 (4 vCPU / 16 GB) — plan using renewal rate	$29	Verified list (~$12.99 intro / ~$28.99 renew)
Supabase	Pro + Large compute (2 vCPU / 8 GB); $25 plan includes $10 Micro credit → compute net ~$100	~$125	Verified list + estimate
Supabase disk / storage overage	DB disk + file storage beyond included	$40–90	Estimate (usage-based)
Supabase egress overage	Must enable transforms + CDN caching; otherwise Normal model (~23,065 GB) is not affordable on Supabase egress rates	$150–400	Estimate if optimized to ~2–6 TB cached
PITR add-on	7 days	$100	Verified list
Redis	Upstash / Redis Cloud free–small	$0–10	Estimate
Monitoring	Uptime + basic error tracking	$0–29	Estimate
Resend / Expo / Places	Existing transactional usage	$20–60	Estimate variable
Budget total (typical)		**~$450–750**	Requires media optimization; unoptimized egress can exceed $1k alone

B. Recommended production

Service	Plan / size	Monthly	Status
Hostinger VPS	KVM 8 or 2× KVM 4	$50–60	Verified (~$49.99 KVM8 renew; or 2×$29)
Supabase	Pro + XL compute (4 vCPU / 16 GB) ≈ $25 + ($210–$10 credit)	~$225	Verified compute table
Realtime overage	Peak connections >500 @ $10 / 1k	$50–150	Estimate from peak concurrent model
Media egress strategy	**Preferred:** Cloudflare R2 (~$0.015/GB-mo, egress free) for Posts bucket **Alt:** Supabase Storage + transforms + Smart CDN cached egress ($0.03/GB overage)	$40–120 (R2 path) or $200–500 (Supabase path)	R2 rates verified; egress volume estimated
PITR 14 days	2× 7-day units	$200	Verified unit price ×2
Redis	Managed 1–2 GB	$15–45	Estimate

Cloudflare	Proxied DNS/WAF for api/site	$0–20	Free tier often enough; Pro optional
Monitoring / logs	APM + alerts	$30–80	Estimate
Email / push / Places	Production volume	$40–100	Estimate
Recommended total (R2 media path)		**~$650–1,000**	Most likely mid ~$800
Recommended total (Supabase media only, optimized)		**~$850–1,400**	Egress swing factor

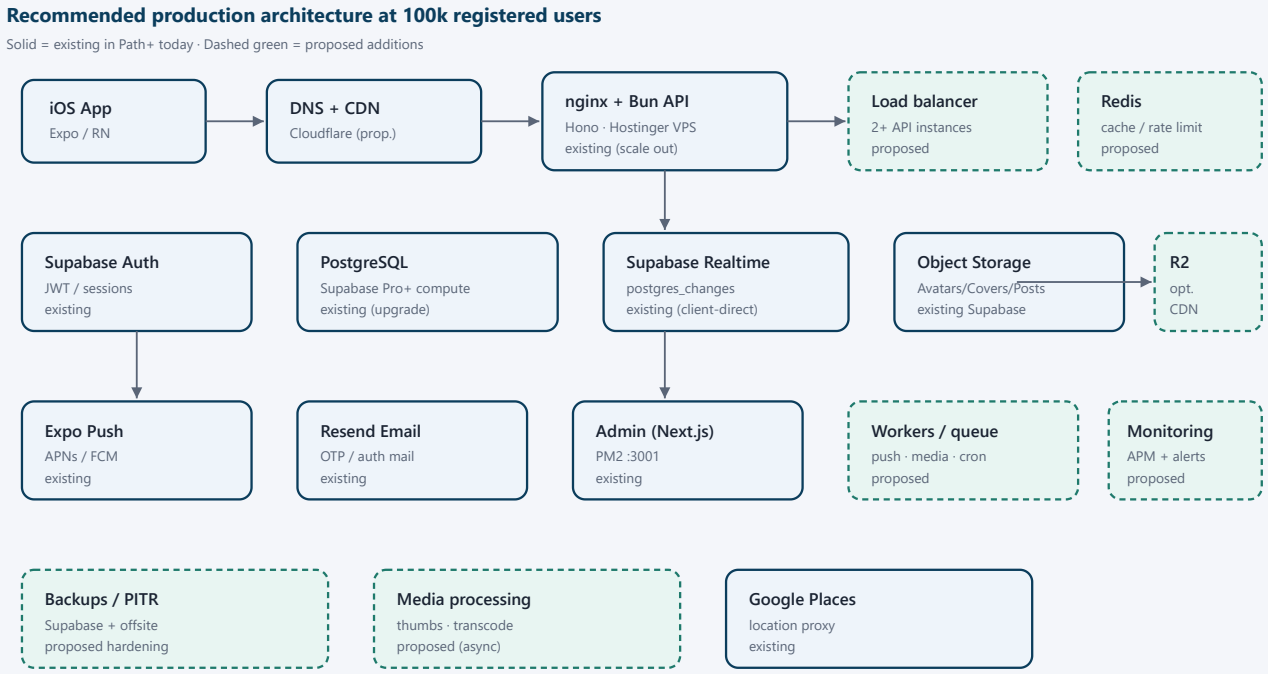
> **Unoptimized warning:** The Normal bandwidth model (~23,065 GB/month) at Supabase uncached egress ($0.09/GB beyond 250 GB) would imply roughly *thousands of dollars per month in egress alone*. That is why image transforms, video preload changes, and/or R2 (zero egress) are mandatory for a 100k-user Path+ — not optional polish.

C. High-availability configuration

Service	Plan / size	Monthly	Status
VPS / compute	3× API-capable hosts or upgraded fleet	$120–220	Estimate on Hostinger list rates
Supabase	Team ($599) + 2XL compute ($410) – credits	~$1,000+	Verified list
R2 media + Cloudflare	1–2 TB storage, free egress	$30–80	Verified storage rate
Redis HA + workers	Production cluster	$60–150	Estimate
PITR + offsite backups	14–28 days + object versioning	$200–350	Mix verified/estimate
Observability	Full APM/log drain	$100–250	Estimate (Supabase log drain $60+)
HA total		**~$1,700–2,600+**	Team plan drives step-up; keep media on R2

> MAU note: Supabase Pro includes 100,000 MAU; if monthly active approaches registered 100k, overage is $0.00325/MAU (verified). Path+ DAU model is lower than MAU; still monitor Auth MAU separately from DAU.

13 Recommended Infrastructure Architecture



Existing vs proposed

- **Keep:** iOS app, nginx, Bun/Hono API, Next admin, Supabase Auth/DB/Realtime/Storage, Expo Push, Resend, Google Places proxy.
- **Upgrade:** Supabase plan/compute; PM2 memory; enable image transforms; protect health/test endpoints.
- **Add:** Redis (rate limit + auth/block caches), background worker, CDN/WAF, monitoring, PITR, optional R2 for egress-sensitive media, load-balanced API instances.

14 Scaling Roadmap

Stage	Users	Infra	Triggers to advance
1	≤10k	KVM 2–4; Supabase Pro Micro/Small; fix P0 rate-limit; remove boot backfills; basic uptime	p95 API >800ms; 429s; DB CPU >60%
2	10–50k	KVM 4; Supabase Medium/Large; enable transforms; signed uploads; Redis; push batching; notification retention	Realtime connection near 500; egress bill spike; feed p95
3	50–100k	KVM 8 or dual VPS; Supabase Large/XL; 2 API instances; workers; CDN/WAF; PITR; query RPCs for feed/inbox	Need HA; MAU near 100k; multi-region latency complaints
4	>100k	HA topology; Team/Enterprise Supabase or read replicas; R2 media; consider message partitioning / cold storage	Sustained Heavy scenario metrics

Avoid premature Kubernetes. Path+'s bottleneck is chatty PostgREST access patterns and media egress—fix those before orchestrating containers.

15 Final Technical Recommendations

1. **Immediately (code):** Fix rate-limit keying; gate boot migrations; remove or auth-protect test/ready leaks; filter realtime message subscriptions.
2. **Immediately (infra):** Move off Supabase Free to Pro; size compute to Large minimum for 100k target planning; enable PITR; raise PM2 memory limits.
3. **Next sprint:** Collapse feed/inbox/badge into SQL RPCs; batch Expo pushes; signed direct uploads; enable image transforms.
4. **Before 50k DAU-equivalent:** Redis + dual API processes; Cloudflare in front; video preload tuning; notification retention job.
5. **Do not:** Host Postgres or user media on the Hostinger VPS; scale instances before fixing shared in-memory caches; treat modeled RPS as live metrics.

VPS TARGET

8 vCPU / 32 GB

or 2× 4 vCPU / 16 GB

DATABASE TARGET

Supabase XL

Pro + 4 vCPU / 16 GB RAM

MEDIA / CDN

R2 + CDN

~$650–1,000 / mo recommended OpEx

16 Sources & References

Primary evidence (inspected)

- Backend tree: `C:\Users\admin\Downloads\new\path-plus-main\path-plus-main\backend` — package.json, ecosystem.config.cjs, .env.example, src/**, deploy/**, prisma stub, migrations
- Mobile tree: `C:\Users\admin\Downloads\path-plus-main\path-plus-main\mobile` — app.config.js, src/lib/**, feed/chat/push hooks

Pricing pages consulted (public)

- https://supabase.com/pricing — Pro/Team, compute sizes, egress, storage, realtime, PITR, MAU
- https://supabase.com/docs/guides/platform/compute-and-disk
- https://www.hostinger.com/vps-hosting — KVM 1/2/4/8 intro & renewal
- https://developers.cloudflare.com/r2/pricing/ — storage & operations; egress free

Not available / requires measurement

- Live VPS CPU/RAM/disk utilization and exact Hostinger SKU
- Supabase project region, current plan tier, row counts, pg_stat_statements
- Production nginx access-log RPS and p95 latencies
- Actual average media object sizes in Storage

> This document is an infrastructure assessment of the **existing** Path+ application. No application source, schema, or production configuration was modified to produce this report.